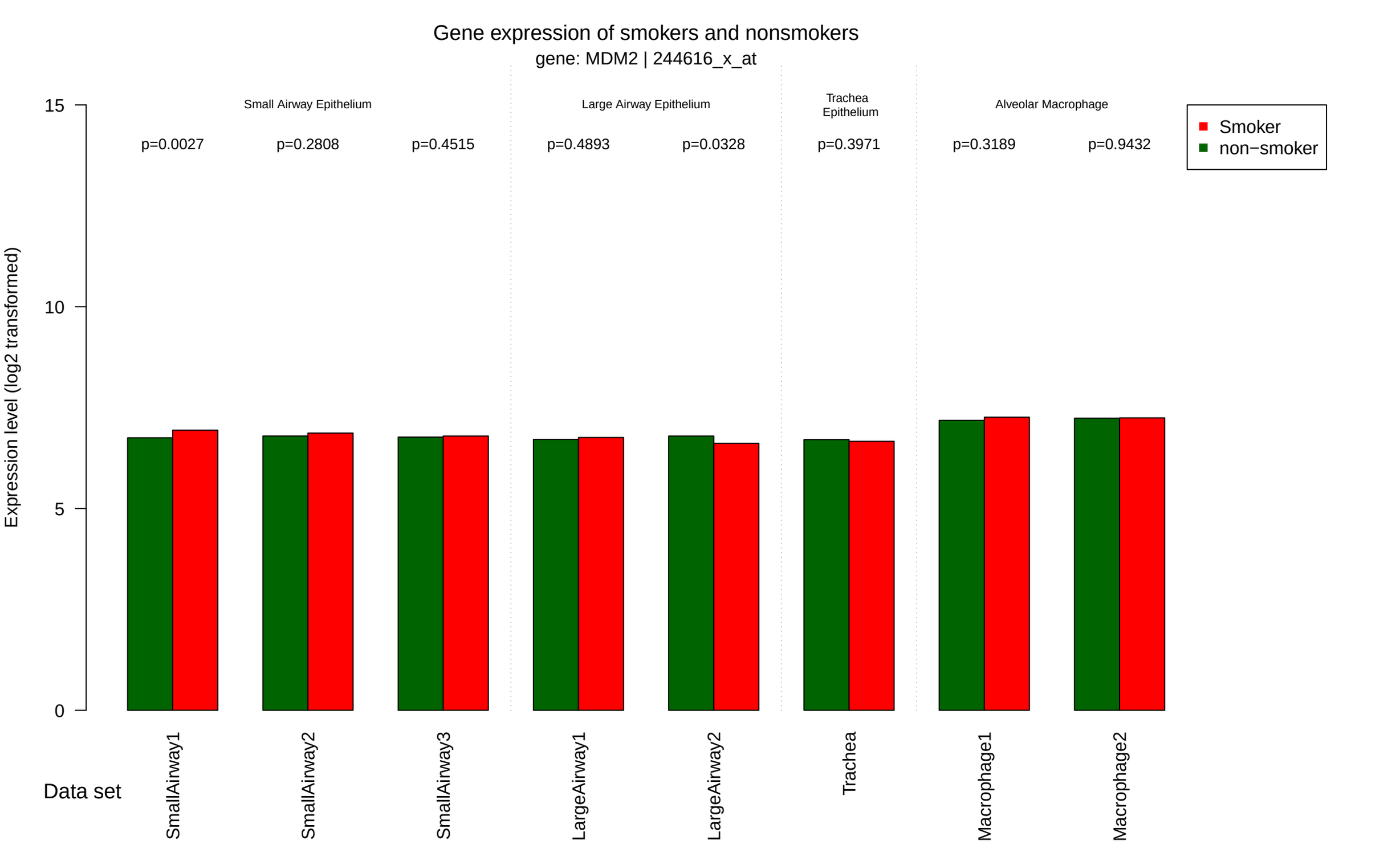
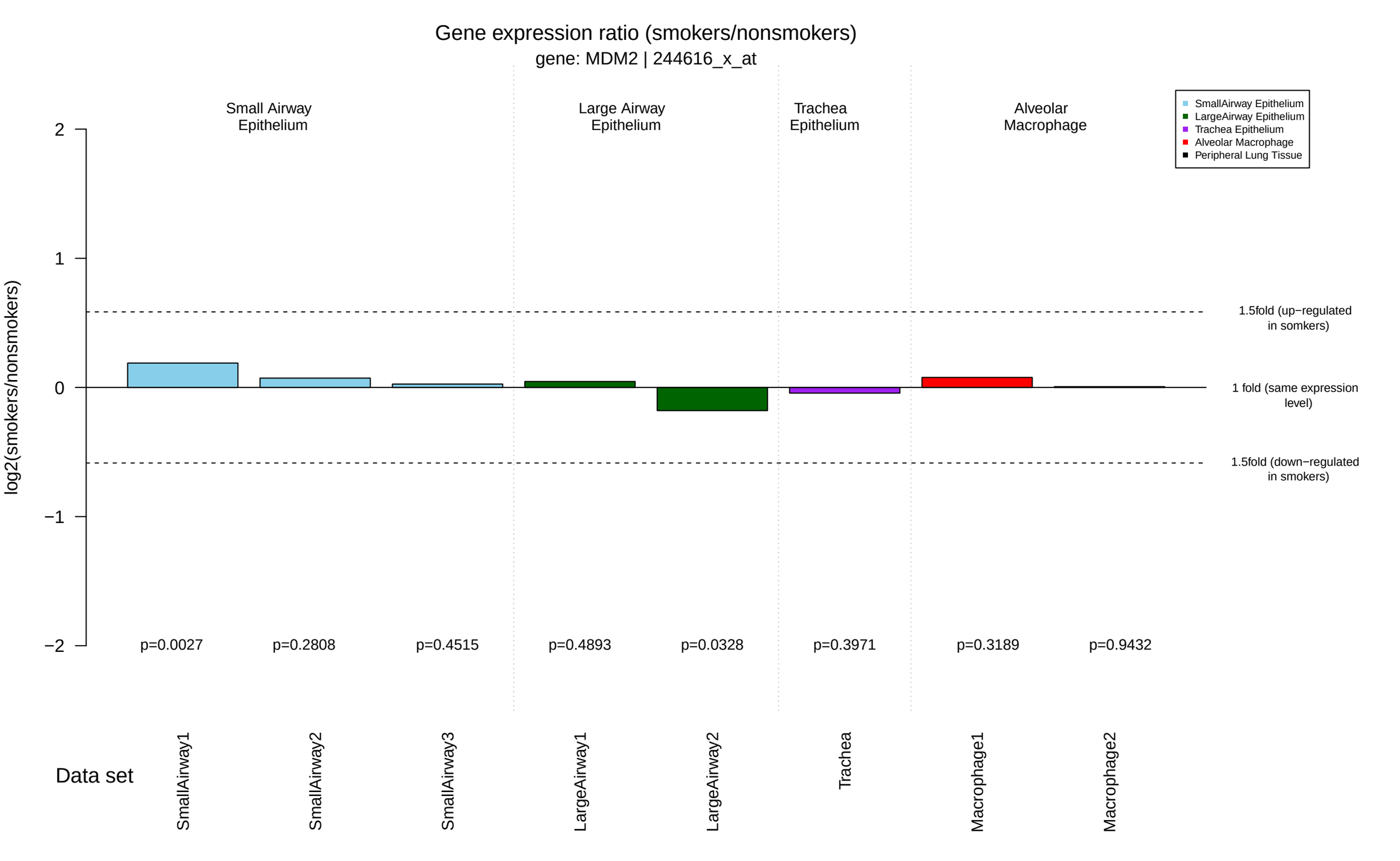
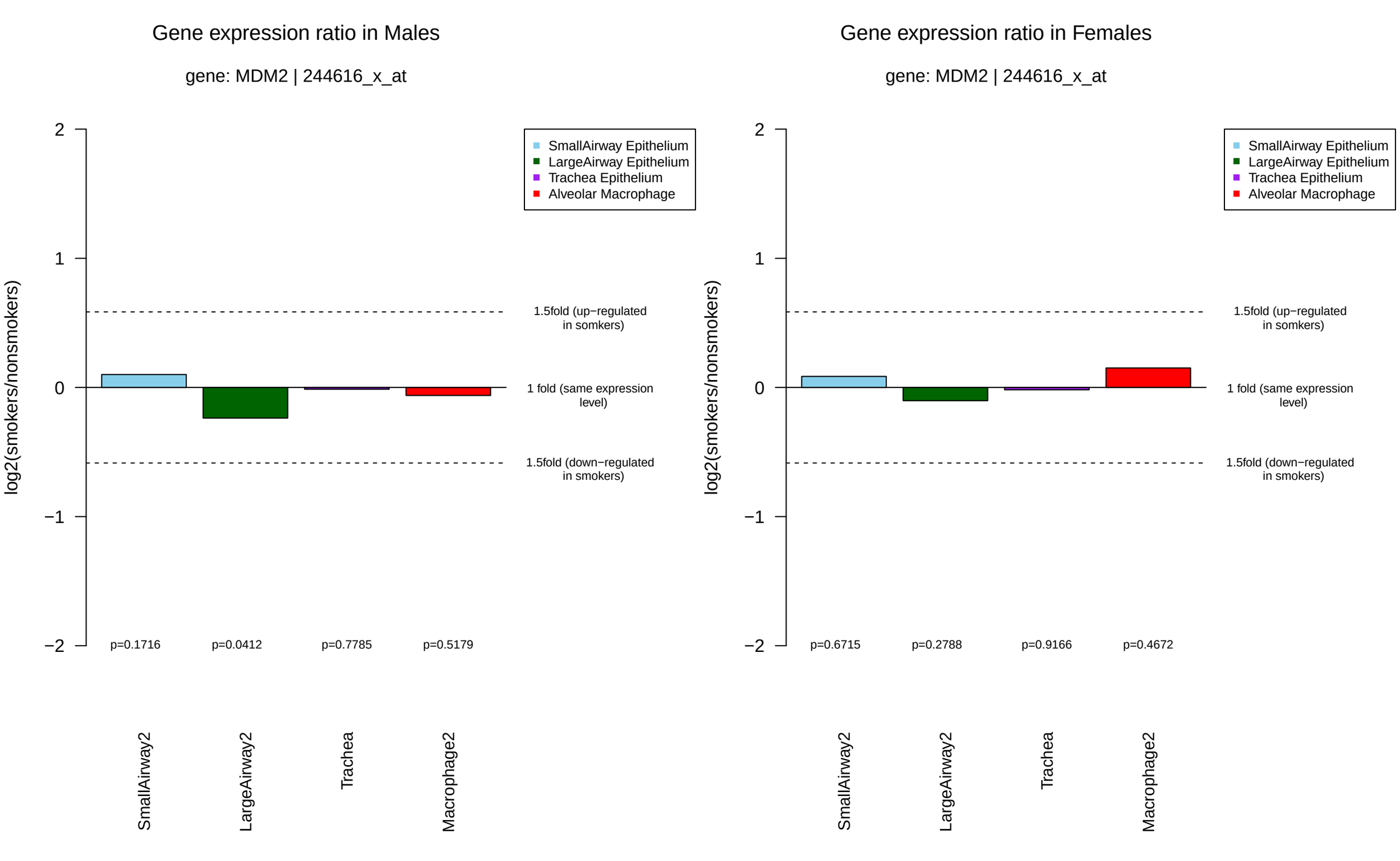
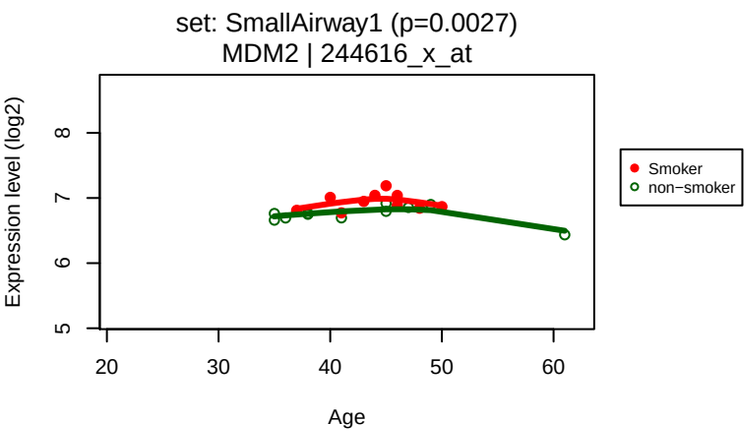
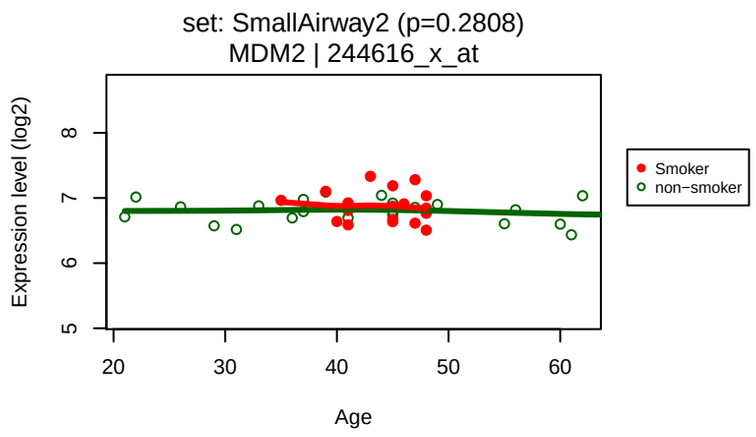
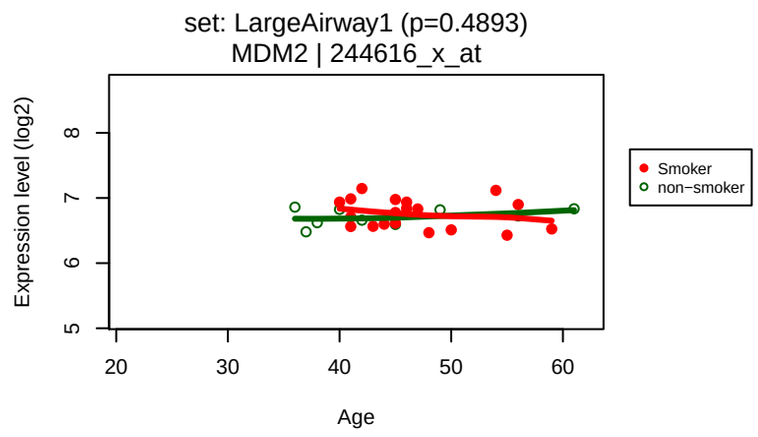

Gene expression of smokers and nonsmokers
gene: MDM2 | 244616_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0027
p=0.2808
p=0.4515
p=0.4893
p=0.0328
p=0.3971
p=0.3189
p=0.9432
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

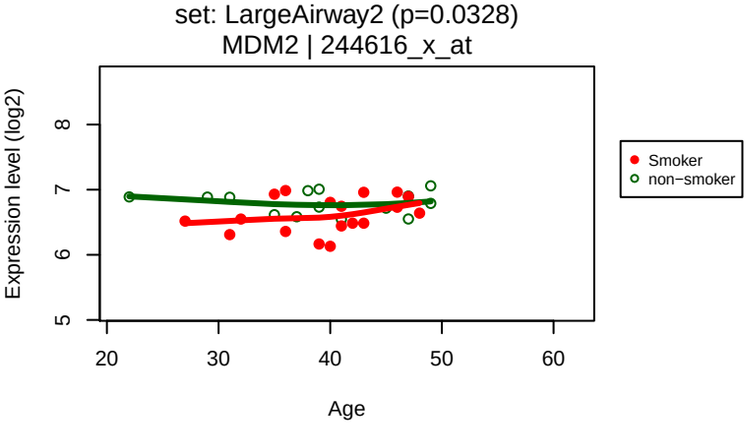

Gene expression ratio (smokers/nonsmokers)
gene: MDM2 | 244616_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0027
p=0.2808
p=0.4515
p=0.4893
p=0.0328
p=0.3971
p=0.3189
p=0.9432
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

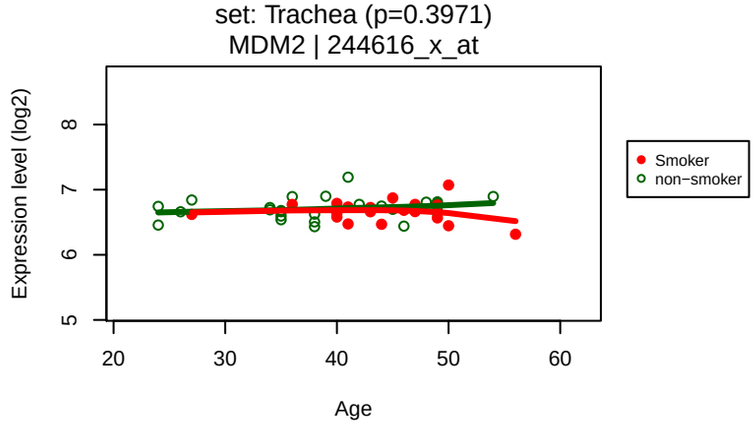

Gene expression ratio in Males
gene: MDM2 | 244616_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1716
p=0.0412
p=0.7785
p=0.5179
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MDM2 | 244616_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6715
p=0.2788
p=0.9166
p=0.4672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

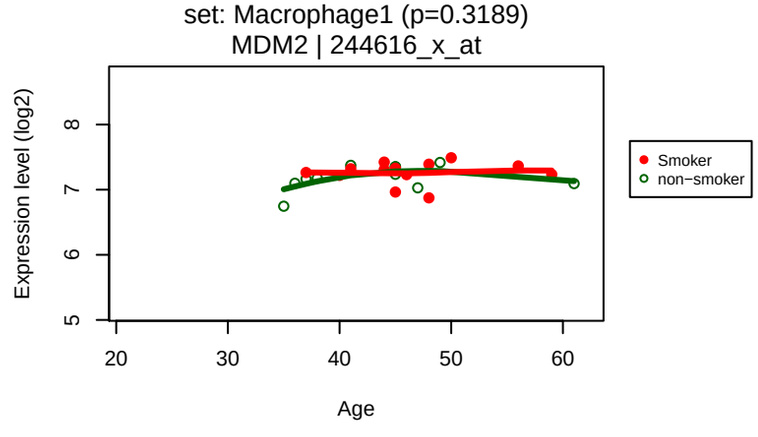

set: SmallAirway1 (p=0.0027)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker

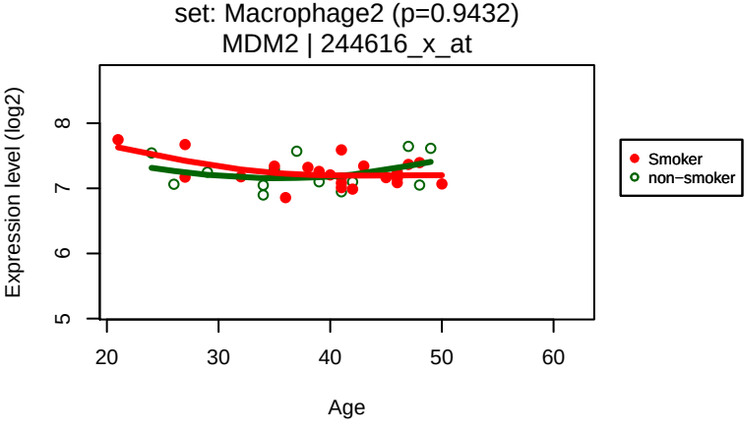

set: SmallAirway2 (p=0.2808)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker

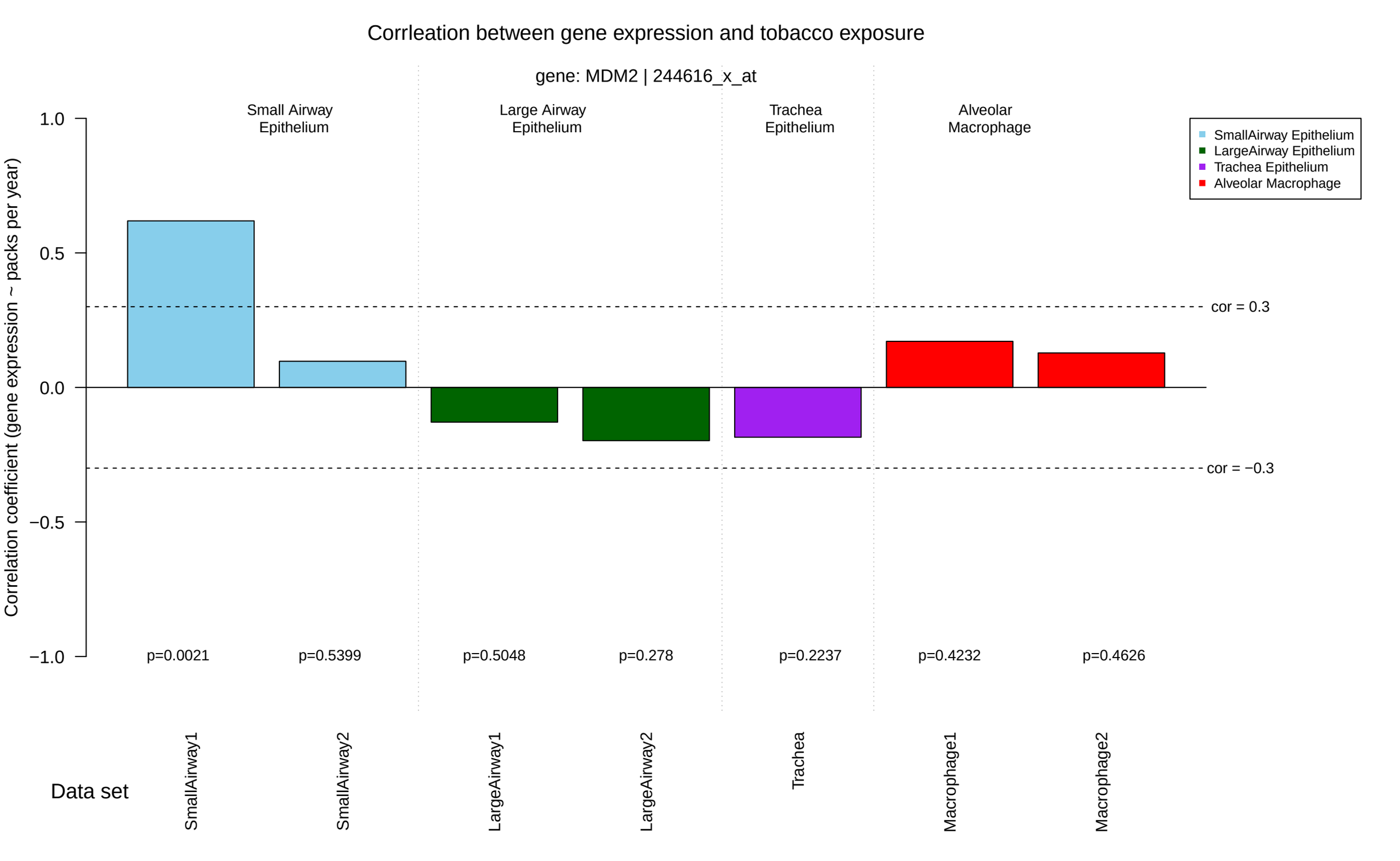

set: LargeAirway1 (p=0.4893)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0328)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3971)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3189)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9432)
MDM2 | 244616_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MDM2 | 244616_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0021
p=0.5399
p=0.5048
p=0.278
p=0.2237
p=0.4232
p=0.4626
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2